EcoVerse

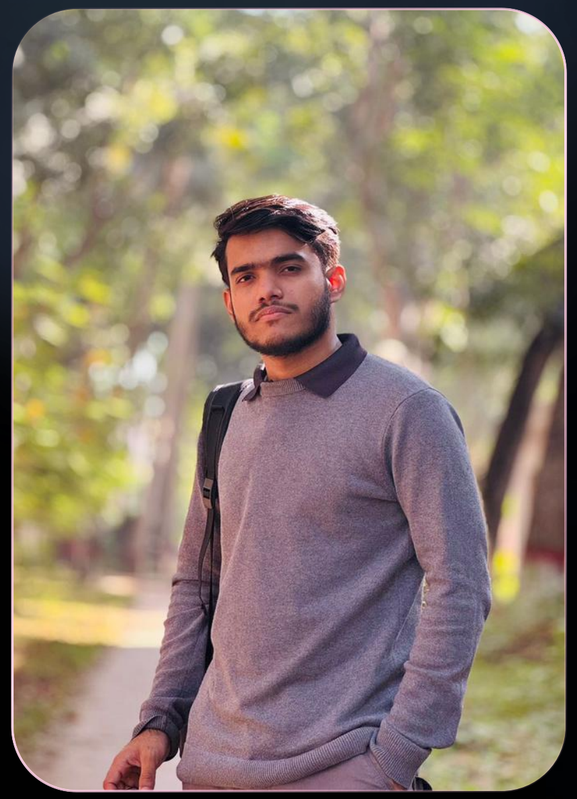
SUSTAINABLE SOLUTIONS FOR A BETTER TOMORROW

HULT PRIZE -2026

Begum Rokeya University, Rangpur

Presented by: Team EcoVerse

Meet Our Team



Abdullah AL Fahad
Founder & CEO

Md Nafiur Hasan
Founder & COO

Al Tabassum Shuva
Founder & CMO

Tasfia Tanzim Neha
Founder & CFO

Did you know that billions of plastic straws are used worldwide every single day?

- United States: About 500 million straws are used daily, making the U.S. a consistent top user accounting for roughly 25-50% of the global total.

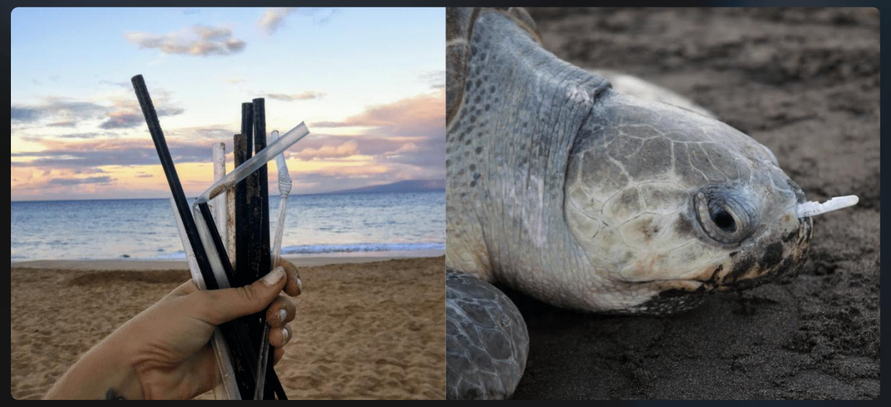
- Asia-Pacific (e.g., China, India): Daily usage ranges between 350-750 million straws, showing the fastest growth, especially in developing markets.

- European Union (especially the UK): Around 250 million straws are used daily; the UK led the EU in implementing pre-ban regulations.

- Malaysia: Usage per capita is very high, with surveys showing that 22% of people use plastic straws daily.

Problem Statement

Plastic straws drive a global pollution crisis. The U.S. discards approximately 500 million daily, contributing to 8.3 billion polluting beaches worldwide annually. Taking up to 200 years to decompose , these non-biodegradable items overwhelm landfills and waterways.

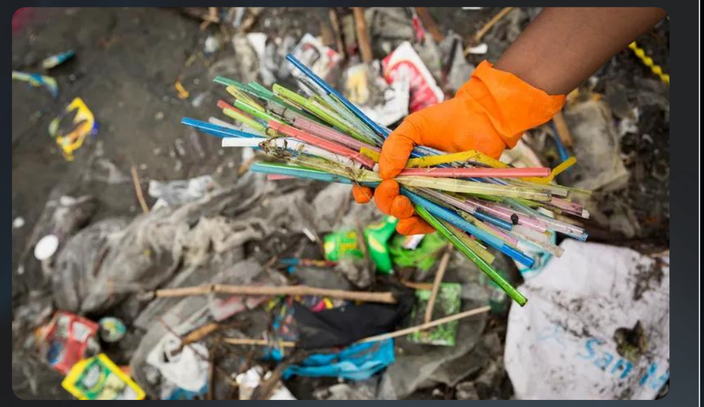


Ocean Threat

8-12 million tons enter marine environments yearly, entangling and killing marine life like turtles and seabirds.

Microplastic Leacjing

Straws break down into microplastics that infiltrate water systems, ecosystems, and human health.



Recycling Failure

<1% recycled; processes release airborne toxins contaminating air, soil, crops, and the food chain.

Solution

Ecoverse introduces , an innovative drinking straw made from Liporonia Grass, a naturally occurring wild grass. It is a 100% natural, biodegradable, and chemical-free alternative to single-use plastic straws.

Unlike plastic straws that can take up to 200-500 years to decompose, it naturally breaks down within 60-90 days, leaving no microplastics, toxins, or harmful residue in the environment.

Mission & Vision

Mission:

Our mission is to reduce plastic pollution by producing affordable, fully biodegradable drinking straws made from sustainably harvested wetland grass promoting environmental conservation, lowering carbon emissions, and empowering rural communities through green livelihoods..

Vision:

By 2030, Ecoverse envisions a planet where single-use plastic straws are no longer needed where sustainable, nature-based solutions dominate the global market and ecosystems thrive without plastic pollution.

Why EcoVerse

Strengthening Local Industry

- *Supports rural farmers and women-led microenterprises*
- *Decentralized processing units create local employment*
- *Reduces dependence on imported or synthetic materials*

Cost-Effective & Sustainable

- *Minimal machinery & low-energy production*
- *Naturally renewable raw material*
- *Lower production cost than bamboo, metal, or plastic alternatives*

Durable & User-Friendly

- *Stronger than paper, does not get soggy quickly*
- *Suitable for hot and cold beverages*
- *Neutral taste and odor-free*

Production Process

Harvesting

Mature Liporonia grass stems are collected from local fields.

Cleaning & Sorting

Grass stems are thoroughly washed to remove dirt, sand, and impurities

Heat Treatment & Sterilization

Stems undergo natural heat treatment or steam sterilization to remove microbes and ensure hygiene.

Drying

Cleaned and treated stems are sun-dried or dried in controlled low-energy chambers.

Cutting & Size Grading

Thickness and diameter are graded to ensure consistency.

Quality Inspection

Each batch is inspected for uniformity, strength, and cleanliness.

Competitor Analysis

Competitor	Material Sustainability	Production Time	Customization Options	Cost Advantage	Eco-Friendliness	Quality Assurance
Plastic	Petroleum-based (non-renewable)	Mass (hours)	High (colors/shapes)	High ($10-15/1K)	Poor (200+ yr decomposition)	Consistent but toxic
Paper	Tree pulp (deforestation risk)	Medium (days)	Medium (print)	Medium ($40-50/1	Fair (months, industrial)	Soggy after 20 min
Bamboo	Renewable bamboo	Long (weeks growth	Low (natural only)	Low ($80-100/1K)	Good (6-12 months)	Durable but woody taste
EcoVerse	Abundant Lepironia grass (fast-growing wetland)	Fast (days harvest)	High (Shape	Strong ($20-30/1K)	Excellent (weeks, home-compostable)	Firm 45+ min, tasteless

SDGs We Meet

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
14 LIFE BELOW WATER
13 CLIMATE ACTION
EcoVerse

Business Model

Business Model

Target Customers

- B2B: Restaurants, cafés, hotels
- B2C: Retail customers

Revenue Streams

- Bulk supply contracts
- Retail pack sales
- Export others countries

Cost Structure

- Raw materials
- Production & labor
- Packaging
- Distribution

Pricing Strategy

- Cost per straw
- Selling price
- Expected profit margin

Market Size

Market Size
$26.1 B

CAGR
6.65%

Financial Breakdown

Metric	Calculation	Amount (USD)
Production Cost	1,000 × $0.01	$10
Revenue (Wholesale)	1,000 × $0.019	$19
Gross Profit	$19 - $10	$9
Gross Margin	$9 ÷ $19	47.4%

EcoVerse 3-Year Projection with Unit Sales (2026-2028)

Year	Units Sold	Revenue (Units × $0.019)	COGS (Units × $0.01)	Gross Profit	OpEx (Optimized)	Net Profit	Net Margin
2026	25.3M	$0.48M	$0.25M	$0.23M	$0.13M	$0.10M	21%
2027	150M	$2.85M	$1.50M	$1.35M	$0.70M	$0.65M	23%
2028	500M	$9.50M	$5.00M	$4.50M	$2.00M	$2.50M	26%

Environmental Impact

1 PLASTIC STRAW → 1.46 G CO_2 EMITTED

1,000,000×1.46 G=1,460 KG CO_2

By replacing 1 million plastic straws with Grass Straws, we can prevent 1,460 kg of CO_2 from entering the atmosphere, protecting our environment and supporting sustainable living.

We invite everyone to join the Ecoverse mission and help save our planet.

Thank You